Manting Guo

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Shanghai Jiao Tong University | Shenzhen Loop Area Institute (Joint Ph.D.)



Education

2024 – Present **Ph.D. Program in Applied Mathematics, Shanghai Jiao Tong University**
Supervisor: Prof. Yan Wang
Joint Ph.D. Program with Shenzhen Loop Area Institute (since September 2025)
Supervisor: Prof. Zizhong Chen
GPA: 3.6/4.0

2020 – 2024 **B.Sc. in Applied Mathematics, Shanghai Jiao Tong University** *(Strengthening Basic Disciplines Plan; graduated with honors).*
GPA: 3.65/4.30 (87.2/100)
Thesis: *Mathematical Modeling and Numerical Simulation of DNA and Protein Aggregation Phenomena.*

2014 – 2020 **Guangzhou Foreign Language School**

Research Experience

2025 – 2026 **Stage-Aware Group Scheduling for Offline Agent Workloads**
Authors: Manting Guo, Yan Wang, Bowen Pang
Independently formulated, implemented, evaluated, and wrote AgentMIP, a DAG-aware mixed-integer scheduler that groups compatible stages across offline agent workflows while accounting for workflow dependencies and adapter switching.
Manuscript under final decision at AAAI 2027.
OpenReview

Branch, Fail, Recover: A Spectral Theory of Reliable Agent Execution
Author: Manting Guo
Analyzed branching, retries, and recovery in agent workflows using spectral methods; derived bounds on execution work and runtime tails, and evaluated the approach through simulations and live-agent experiments.
Under review at AISTATS 2027.

Beyond Sequence Metrics: State-Space Coverage and Algorithmic Design in Deterministic Channel Hopping
Modeled deterministic channel hopping with state graphs, designed and ran simulations, and contributed to writing the manuscript.
Under review.

On Some Problems of Subcube Partitions
Authors: Manting Guo and Peiru Kuang (equal contribution)
Established asymptotic enumeration results and characterized typical structural properties of subcube partitions.
arXiv preprint.

Research Experience (continued)

2023 – 2024 **Mathematical Modeling and Numerical Analysis of Reaction-Diffusion Systems**
Developed the theoretical framework and carried out the theoretical derivation, numerical simulations, and analysis for a study of biomolecular condensates and mass-conserving reaction-diffusion systems.
Related article: "Biomolecular condensates bridge experiment and theory of mass-conserving reaction-diffusion systems in phase separation," Structure*, 2025.*
Article

2020 – 2024 **Undergraduate Research Projects**
Cross-scale systems: Derived scaling laws and performed bifurcation analysis using Fortran and AUTO.
Arranged forests: Explored structured subset-selection problems and a prime-based construction related to reducing feature overlap between random-forest trees.
These projects resulted in research reports.

Engineering and Open-Source Contributions

CANN ops-blas — Contributing Arch22 support for aclblasCsbmv (merge request #681, in progress).

384-supernode platform — Tested the Lusso scheduler across communication, training, and inference-service workloads. Separately tested HCCL, the Huawei Collective Communication Library, and investigated test failures.

MCRD to JAX — Porting the DNA–p53 MCRD simulation code to JAX; validation and performance testing are in progress.

Publications

Published

2025 **Biomolecular condensates bridge experiment and theory of mass-conserving reaction-diffusion systems in phase separation**
Cheng Li, Man-Ting Guo, Xiaoqing He, Quan-Xing Liu, and Zhi Qi.
Structure, 33 (2025), 1–14. Article

Preprint

arXiv **On Some Problems of Subcube Partitions**
Manting Guo and Peiru Kuang (equal contribution).
arXiv preprint.

Under Review

AAAI 2027 **Stage-Aware Group Scheduling for Offline Agent Workloads**
Man-Ting Guo, Yan Wang, and Bowen Pang.
Under final decision. OpenReview

AISTATS 2027 **Branch, Fail, Recover: A Spectral Theory of Reliable Agent Execution**
Manting Guo.
Under review.

Under review **Beyond Sequence Metrics: State-Space Coverage and Algorithmic Design in Deterministic Channel Hopping**
Manting Guo, Peiru Kuang, and Yan Wang.
Under review.

Skills and Interests

Coding	Python, Matlab, C++ & C, R, Fortran (basic), LaTeX.
Languages	English: CET4:642 CET6:598 IELTS:7.0 tofel: 5.5 / 6.0 German : A2
Academic interests	My main research interest lies in optimization and infrastructure for AI-agent workflows. More broadly, I am drawn to mathematically grounded problems whose impact can be observed in real-world systems, from biological pattern formation and scientific computing to future applications in AI-driven decision-making, finance, education, and other complex domains. Curious about diverse subjects.
Other interests	Fitness and weight training, guitar (Fingerstyle), model building and assembly ,ball games, hiking, …

Miscellaneous Experience

Awards and Achievements

2026	**Outstanding Academic Scholarship, School of Mathematical Sciences**
2025	**Outstanding Academic Scholarship, School of Mathematical Sciences**
2024	**School level outstanding graduate.**
	Department Prize for Outstanding Bachelor thesis
2023	**Third-Class Scholarship, School of Mathematical Sciences**
2022	**Outstanding Academic Scholarship, School of Mathematical Sciences**
2021	**Outstanding Academic Scholarship, School of Mathematical Sciences**
	School level Basketball competition, bronze medalist.
2020	**School level Basketball competition, Second place.**

Misc.

2021-2024	**Served as a member of the Student Affairs Center's General Affairs Department for two years, with one year as the core team member.**
2022	**Served as the group leader of the Return-to-School Presentation project at Shanghai Jiao Tong University, responsible for organizing and leading the group's activities.**
2020-2024	**Served as Class Monitor during undergraduate studies.**